INT421—Digital Signal Processing

An Introduction and Course Logistics

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Faculty of Information Technology and Computer Sciences





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1. Welcome to INT421—Digital Signal Processing
2. Course Polices
3. What is Signal, & What is Signal Processing?
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5. What We Learn in INT421—Digital Signal Processing
6. Application Areas of DSP
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General Course Information

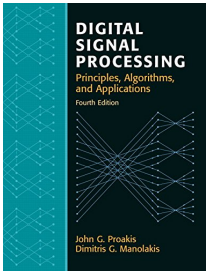
- Instructor
  - Dr. Mohammed El-Telbany
- Course Teaching Assistant
  - Asmaaa Awad
- Location
  - Hall no. N203
- Class Times
  - Tuesday from 12:20 pm to 2:00 pm
- Email
  - mohammed.elsaid@su.edu.eg
- Office Hours
  - Tuesday evening 3:00-4:00 PM, for consulting, advising, and problem solving





Course Literature

- Prerequisites
  - MA110—Linear Algebra
  - MA111—Calculus.
- Credit Hours
  - (2+0+2)
- Text Book(s)
  - J. Proakis and D. Manolakis (2007). *Digital Signal Processing: Principles, Algorithms, and Applications*, 4th Edition, Pearson.
- References
  - J. McClellan, R. Schafer and M. Yoder (2016). *DSP First*, 2nd Edition, Pearson.
- Course Website
  - https://kmoodle.su.edu.eg/



Other Support Sources

- Supplemental Textbook(s)
  1. *DSP First*, 2nd edition, by J. McClellan, R. Schafer and M. Yoder, Pearson, 2016.
  2. *Discrete-Time Signal Processing*, 3rd edition, by A.V. Oppenheim and R.W. Schafer, Pearson, 2009.
  3. *Applied Digital Signal Processing Theory and Practice*, 1st edition, by D.G. Manolakis, V.K. Ingle, Cambridge, 2011.
- Office Hours
  - Three hours per week for instructor and TA.
  - You can schedule additional times with me as needed.
- The LMS Course Page
  1. Lecture slides
  2. Lecture Videos
  3. Handouts
  4. Exercises/Assignments.

Academic Stress and Sources of Help

- This is a hard course
- Take good care of yourself
- If you are having trouble, seek help
  1. Teaching staff
- We are here to help!

Course Objectives

This course will help students to achieve the following objectives:

1. Understand analog vs. digital signal representation and processing.
2. Understand the basic properties of time-invariant linear systems;
3. Understand sampling, aliasing, convolution; filtering, the pitfalls of spectral estimation;
4. Explain the above in time and frequency domain representations;
5. Visualize and discuss digital filters in the Z-domain;
6. Use the FFT for convolution, deconvolution, filtering
7. Implement, apply and evaluate simple DSP applications in MATLAB.

Course Outline

The following is an outline for the course, subject to alteration as the semester proceeds.

1. Overview of DSP.
2. Signals, systems, and Frequency Domain Analysis.
3. Discrete-time systems.
4. Z-transform and transfer function.
5. Architectural design.
6. Fourier series and Fourier transform.
7. Basic filter structure and theory.
8. Design of Digital Filters.

Team Projects

- Typical projects have either a software/hardware implementation component and an oral presentation.
- Past projects include:
  - Compression algorithms
  - Speech enhancement
  - Filtering a signal for noise reduction.
  - Audio watermarking
- Please feel free to come and talk to me about your ideas for a possible project.

Laboratory Activities

1. The course has an associated laboratory, which will require the use of MATLAB or SCIENTIFIC PYTHON as recommended tools for the class.
2. Lab activity will be use apply MATLAB in experiments such as:
   - MATLAB for Digital Signal Processing.
   - Generation of Continuous/Discrete Time Signals
   - Basic Operations on Discrete Signals
   - Sampling Theorem
   - Discrete Time Systems
   - Discrete-Time Convolution
   - Z-Transform
   - Discrete Fourier Transform
   - Digital Filter Design
   - DSP Applications
3. Student must read / do Post-Lab before come to the laboratory.
4. Lab will be a lab assessment/lab team-project.

Assessment Activities

1. Students will be assessed using exams, quizzes, homework assignments and/or project work typically distribution as follows:

Evaluation item	**Percentage**
Homework and Project	10%
Laboratory	10%
Midterm Exam	20%
Final Exam	60%.

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Exam Policy

- Exams will be a combination of the materials presented in the *lectures* and *homework problems*.
- Missed tests cannot be made up
  - There will be **no makeup exams**; students must instead make arrangements with the instructor to take the exam prior to the planned absence.
  - Consequently, no makeup for missed quizzes.

Attendance/Absence Policy

- The University requires that I take attendance. Class attendance will be taken in each lecture.
- Attend all classes
- Experience shows
  - Poor attendance = Poor performance
- Availability of class lecture slides/videos is no substitute for class attendance.
- Students that exceed class absences (25% of lectures) are being considered dropped from the course and prevented from entering the final exam.

Academic Dishonesty

- The student-teacher relationship is based on trust.
  -
  - Your name on the submitted work is an affirmation that the work is entirely your own.
  - When submitting team work, all team members' names must appear on the work.
- Any student engaging in academic dishonesty, or assisting another student in academic dishonesty.
  - Will receive a grade of **F** for the course.
  - Reported to Dean.
- Academic Dishonesty
  - Copying another student's work.
  - Submitting the work of another as your own.
  - Allowing others to copy your work.

Your Feedback Are Important

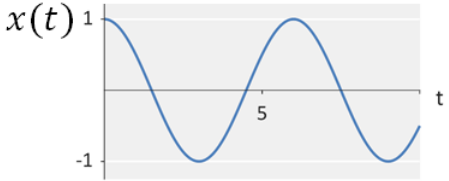
- The technologies keep evolving, and the course materials need to be updated correspondingly
- Please advise where I can improve after each lecturer, at the discussion.
- Online class evaluations will be available for students to complete during the last week of class.
- All evaluations are confidential; instructors will never know how any one student responded to any question, and students will never know the ratings for any particular instructors.
- Course Evaluation Website/Form will be announced later:

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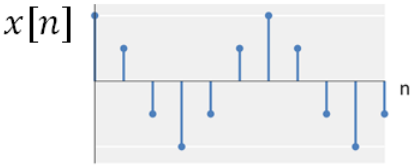
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What is Signal?

- A signal is a physical quantity, or quality, which conveys information.
- A signal is anything you can see, hear, observe or measure using some machine.
- Examples: Speech, audio, light, radio, TV, radar, supersonic, temperature, EEG, etc.



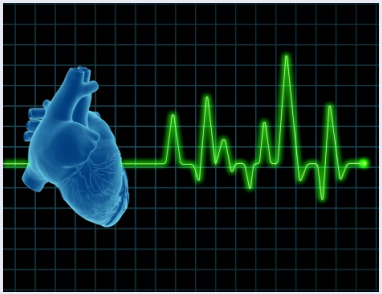
Continuous-time signals



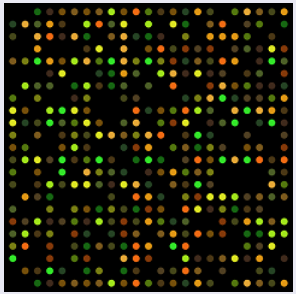
Discrete-time signals

What is Signal?

1D-Biological Signals: Electrocardiogram (ECG)



2D-Biological Signals: Micro-arrays



What is the Meaning of a Signal?

- A signal is the carrier of information.
  - ECG contains information related to sickness.
    - Medical diagnosis
- Voice contains information about who is speaking and what is spoken
  - Speech recognition
  - Speaker recognition
- Radar signal contains information about the target.
  - To avoid accidents for airport navigation
  - To defend the country from enemy's attack
- Biometric images contain information about authentication
  - Finger prints

Examples of Signals

- Signals are everywhere and may reflect countless measurements of some physical quantity such as:
  - electric voltages
  - brain signals
  - heart rates
  - temperatures
  - image luminance
  - investment prices
  - vehicle speeds
  - seismic activity
  - human speech

Signal Dimensions

- **1-D**
  - (e.g. dependent on time).
- **2-D**
  - (e.g. images dependent on two coordinates in a plane (Spatial))
- **3-D**
  - (e.g. describing an object in space)

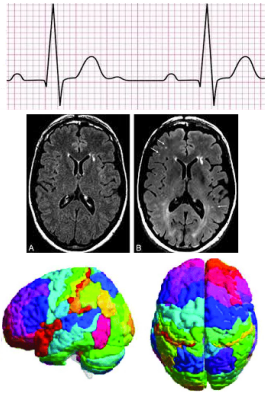


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What is Digital Signal Processing (DSP)?

- **Digital Signal processing** is concerned with the mathematical representation of the signal and the algorithmic operation carried out on it to extract the information present.
- It Consists of three words.
  - **Digital**, **Signal** and **Processing**.

Signal: any (physical or non-physical) quantity that varies with time, space, or other independent variable(s).

Digital: a discrete-time and discrete-valued signal, i.e. digitization involves both sampling and quantization.

Processing: operations on the signal.

What is Digital Signal Processing (DSP)?

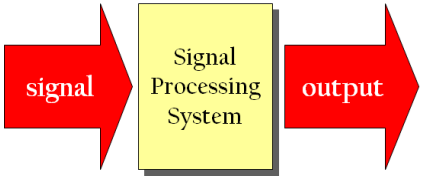
- For most purposes of description and analysis, a signal can be defined simply as a mathematical function,

$$y = f(x)$$

- where x is the independent variable which specifies the domain of the signal e.g.:
  - $y(t) = \sin(\omega t)$ is a function of a variable in the **time domain** and is thus a time signal;
  - $X(\omega) = 1/(-m\omega^2 + ic\omega + k)$ is a **frequency domain** signal;
  - An image $I(x, y)$ is in the **spatial domain**

What is Digital Signal Processing (DSP)?

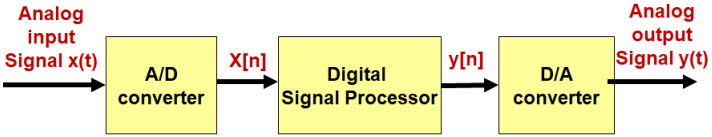
- A **system** A physical entity that operates on a set of primary signals (the inputs) to produce a corresponding set of resultant signals (the output).



- The conversion from excitation to response is called **signal processing**.
- The operations, or processing, may take several forms: modification, combination, decomposition, filtering, extraction of parameters, etc.

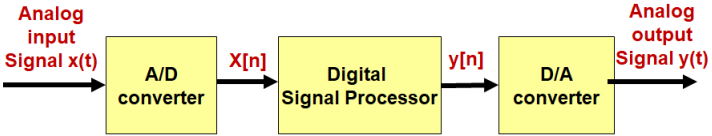
Basic Components of a DSP System

- In its most general form, a DSP system will consist of three main components, as illustrated below:



- The A/D converter transforms the analog signal $x(t)$ at the system input into a digital signal $x[n]$.
- The digital system performs the desired operations on the digital signal $x[n]$ and produces a corresponding output $y[n]$ also in digital form.
- The D/A converter transforms the digital output $y[n]$ into an analog signal $y(t)$ suitable for interfacing with the outside world.

Basic Components of a DSP System



- Analog input – analog output
  - Digital recording of music
- Analog input – digital output
  - Touch tone phone dialing
- Digital input – analog output
  - Text to speech
- Digital input – digital output
  - Compression of a file on computer

Limitations of Analog Signal Processing

- Accuracy limitations due to
  - Component tolerances
  - Undesired solitariness
- Limited repeatability due to
  - Tolerances
  - Changes in environmental conditions.
- Sensitivity to electrical noise
- Limited dynamic range for voltage and currents
- Difficulty of storing information

Advantages of Digital Signal Processing

- **Flexibility**—Digital implementation allows flexibility to reconfigure the DSP operations by simply changing the program.
- **Accuracy**—DSP provides any desirable accuracy by simply increasing the number of bits (word length), while tolerance limits have to be met in the analog counterpart.
- **Easy Storage**—Digital signals can be easily saved on storing media, such as magnetic tape, disk, and optical disk without loss of information. They can also be easily transported and processed off-line in remote laboratories.
- **Processing**—DSP allows for the implementation of more sophisticated signal processors than its analog counterparts do.

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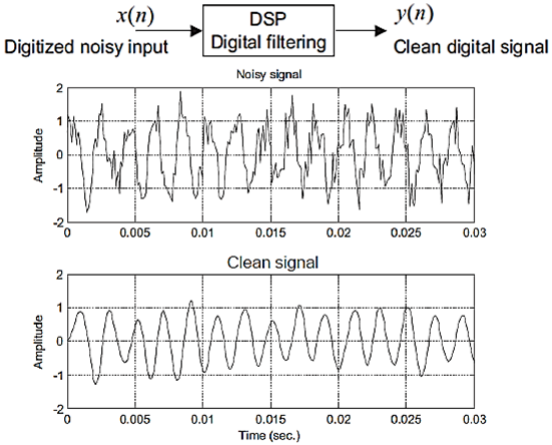
What We Learn in INT421—Digital Signal Processing

- The purpose of signal processing is to make it easier to extract useful information and to make correct decisions. For this purpose, we have
  1. Signal enhancement
  2. Signal restoration
  3. Signal reconstruction
  4. Signal synthesis
  5. Signal estimation and prediction
- After signal processing, we can extract more important information, and recognize, aware, and make decisions.



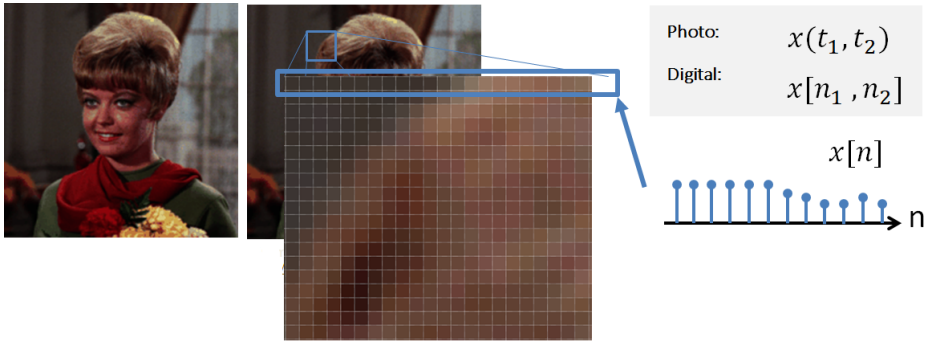
Signal Enhancement

- **Digital Filtering**: The (frequency) filter removes those frequencies in the signal that correspond to noise.



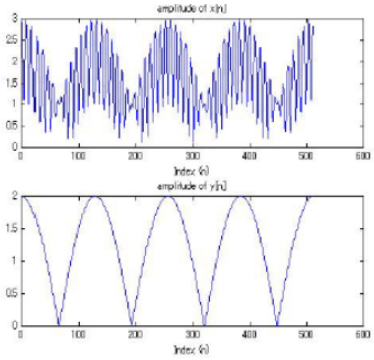
Signal Enhancement

- **Signal amplification** in radio, TV, human ears
- **Image enhancement** in image processing



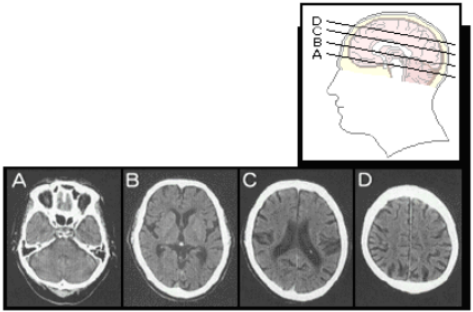
Signal Restoration

- **Restore a signal corrupted** by noise or useless background party
  - Tuning in radio and TV
  - **Compensate** for sensor deficiencies.
  - Pick up the voice of a certain person in a party



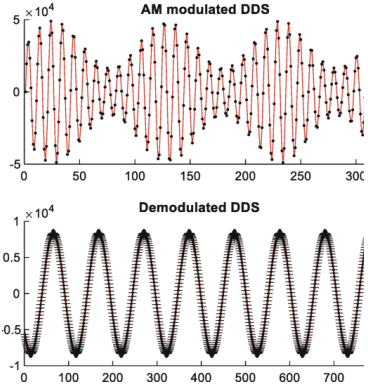
Signal Reconstruction

- **Reconstruct** a signal from different measurement
  - Computerized tomogram (CT).
  - Synthetic aperture radar (SAR) imaging.
  - Reconstruct 3D images from different 2D views.



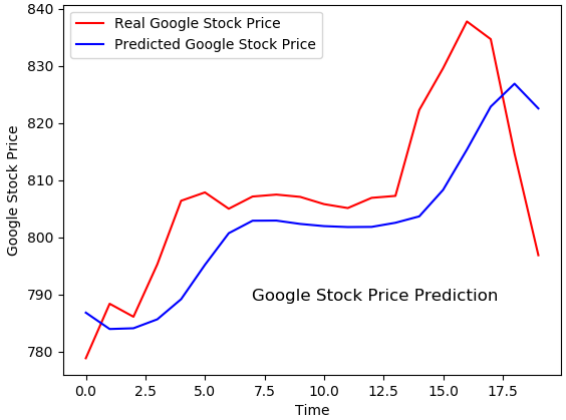
Signal Synthesis

- Find a system that can **generate the desired signal**
  - Speech synthesis
  - 3D sound synthesis
  - Make a virtual world in computer



Signal Estimation

- **Estimate** the future values from observed data
  - Stock prediction
  - Electrical Demand prediction
  - Image compression



Various Types of Processing

1. Modulation and demodulation.
2. Signal security.
3. Encryption and decryption.
4. Multiplexing and de-multiplexing.
5. Data compression.
6. Signal denoising.
7. Filtering for noise reduction.
8. Speaker/system identification.
9. Signal enhancement – equalization. Audio processing.
10. Image processing – image denoising, enhancement
11. Watermarking.
12. Reconstruction
13. Signal generation - music synthesis
14. Data analysis and feature extraction.
15. Frequency/spectral analysis.

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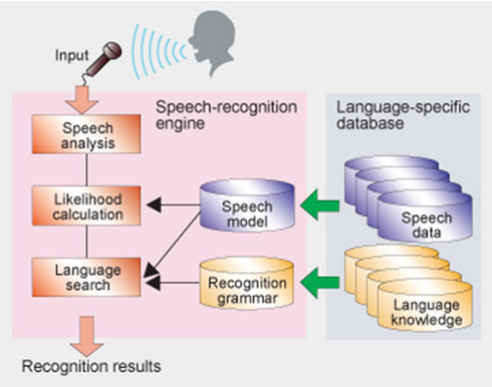
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Application Areas of DSP

- **Speech Processing**—Speech compression and decompression for voice storage system and for transmission and reception of voice signals; speech synthesis in message warning systems.
- **Communication**—Elimination of noise by filtering and echo cancellation, adaptive filtering in transmission channels.
- **Biomedical**—Spectrum analysis of ECG signals to identify various disorders in the heart and spectrum analysis of EEG signals to study the malfunctions or disorders in the brain.
- **Consumer Electronics**—Music synthesis and digital audio and video.
- **Image Processing**—Two-dimensional filtering on images for image enhancement, fingerprint matching, image compression, medical imaging, identifying hidden images in the signals received by radars, etc.
- **Navigation**—Global positioning system (GPS) satellite signal processing for air, sea, and land navigation.

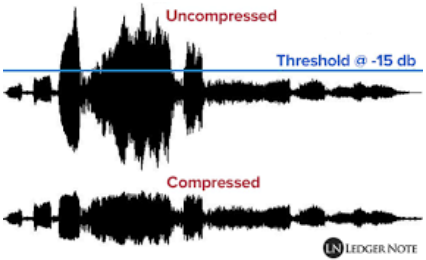
Example 1: Speech Recognition

- **Speech recognition**: DSP generally approaches the problem of voice recognition in two steps, feature extraction followed by feature matching.



Example 2: Audio Compression

- **Compress audio** by 10x without perceptual loss of quality
- Sophisticated processing based on models of human perception.
  - 3MB files instead of 30MB



Example 3: Biometrics

- The "**Biometrics**" field focuses on methods for uniquely identifying humans using one or more of their intrinsic physical or behavioural traits.
  - Examples include using face, voice, fingerprints, iris, handwriting or the method of walking



Example 4: Brain Computer Interface

- A means for communication between a brain and a computer via measurements associated with brain activity.
- No muscle motion is involved (e.g., eye movement).

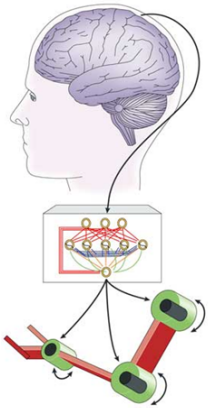


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Summary

- Signal processing: is "**The art of getting what you want from signals**."
- Reviewed basic concepts and terminologies of DSP.
- Examined some practical examples.

Any Questions?

